

Fig. 3: Mobile phone block diagram

a co-processor for different functions. The MCU receives signals from analogue sensors (temperature sensors like thermistors, resistance temperature detectors and humidity sensors that receive analogue inputs and provide digital voltage, which is applied to the MCU), 2/3-axis accelerometers (that measure 2/3-axis movements and convert it to digital voltage, which is applied to the MCU) and ambient light sensors (ALSes) interface.

Ambient light sensor enables automatic control of display backlight brightness over a wide range of illumination conditions, from a dark environment to direct sunlight. With ALS input, an MCU or baseband processor increases or decreases display brightness, depending on the environment. The MCU also receives magnetic sensor inputs through external ADCs and buffer circuitry. Besides, it uses the accelerometer and mechanical joystick for running gaming applications.

The MCU uses proximity sensors for face detection and hand-movement (grip detection) in mobile phones. When you move the phone near your face, the MCU switches off the keypad, and when you move your hand near the keypad, it activates/switches on the keypad. The MCU uses host interrupt feature to

activate broadband and application processor for face detection.

The MCU enables face detection by detecting proximity of your face, ear or head to eliminate false touches of the touchscreen. This reduces talktime power by powering down the touchscreen. It is implemented by using infrared proximity sensors.

The MCU uses a proximity sensor for specific absorption rate (SAR) regulation in tablet applications. SAR is the rate at which energy is absorbed by the human body when exposed to radio-frequency electromagnetic waves. The proximity sensor dynamically regulates radio transmissions near human beings.

The MCU uses external buffer-

driver circuitry to drive LCD or graphical displays. Typically, an LCD or a graphical display with backlight is used in low-end mobile phone applications. The MCU also controls the backlight of LCD and graphical displays. It uses touchscreen controllers for the touchscreen interface in high-end mobile phone applications.

The MCU uses a mechanical keypad for your inputs in low-end mobile phone applications. It uses CapSense/touchscreen based keypads in high-end mobile phone applications.

The MCU monitors lithium-ion battery voltage and is also used for battery-charging application. It is also used for fault detection and data recovery in mobile phones.

The MCU uses haptics (tactile feedback technology) in mobile phones for replacing mechanical buttons that interface with you through the sense of touch by applying force, vibration or motion.

The MCU interfaces with broadband and application processor and other onboard peripherals (like camera interface controller) through I2C and SPI interfaces. It interfaces with broadband and application processor in host (master) and slave mode for data transfer. Broadband and application processor use onboard USB 2.0/3.0 controller to transfer data externally.

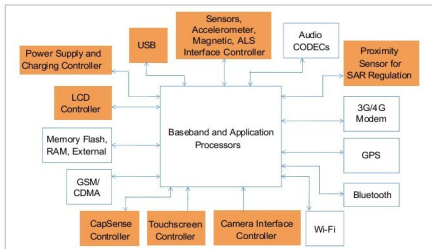


Fig. 4: Tablet block diagram